WEEK 3 · LESSON 15 OF 24 · HANDOUT

Describing Data: Averages and Spread

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=kn83BA7cRNM — open the lesson online to watch inside the course.

Your goal in this lesson: Calculate and interpret the mean, median, mode, range, variance, standard deviation and percentiles, and choose the right summary when data has outliers.

Two questions about any column of numbers

1. **Where is the middle?** (centre)
2. **How spread out are the values?** (spread)

The centre: mean, median, mode

```
import numpy as np
import pandas as pd

# monthly income (RWF) of 9 workers in a small company
income = np.array([180000, 200000, 210000, 220000, 230000,
                   240000, 250000, 260000, 2500000])
print(np.mean(income))     # about 476,667
print(np.median(income))   # 230,000
```

- **Mean** — add all values and divide by how many. Sensitive to extreme values.
- **Median** — the middle value after sorting. Not affected by one extreme value.
- **Mode** — the most common value. Useful for categories (most common product, most common shoe size).

The manager's 2,500,000 RWF pulls the mean far above what a typical worker earns. **When there are outliers, the median describes the "typical" value better.** That is why reports on household income usually quote the median.

The spread: range, variance, standard deviation

Two classes can have the same average mark but very different spreads:

```
class_a = np.array([60, 62, 65, 65, 68, 70])
class_b = np.array([30, 45, 65, 70, 85, 95])
for name, marks in [("A", class_a), ("B", class_b)]:
    print(name, marks.mean().round(1), marks.max() - marks.min(), marks.std(ddof=1).round(1))
```

- **Range** — max − min. Simple, but depends on just two values.
- **Variance** — the average of squared distances from the mean.
- **Standard deviation (SD)** — the square root of the variance, back in the original units. Roughly: *"how far a typical value is from the mean."*

`ddof=1` gives the **sample** standard deviation (divide by n − 1), which is what Pandas uses by default. Use it when your data is a sample from a bigger population.

Percentiles and the five-number summary

The **p-th percentile** is the value below which p% of the data falls. The 25th, 50th and 75th percentiles are the **quartiles**.

```
marks = pd.Series([45, 52, 58, 61, 64, 66, 70, 73, 78, 85, 92])
print(marks.describe())
q1, q3 = marks.quantile([0.25, 0.75])
print("IQR:", q3 - q1)
```

The **interquartile range (IQR)** = Q3 − Q1 is the spread of the middle half. A common rule flags values below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR as possible **outliers**.

Why AI needs these

- **Data cleaning:** spotting outliers and impossible values.
- **Feature scaling:** many models work better when features are rescaled using mean and SD (**standardisation**, next lesson's z-scores).
- **Evaluation:** a model's average error is a mean; its consistency is a spread.
- **Honest reporting:** "average" without spread can mislead.

Key terms

- **Mean** — the arithmetic average.
- **Median** — the middle value of sorted data.
- **Mode** — the most frequent value.
- **Standard deviation** — the typical distance of values from the mean.
- **Percentile / quartile** — value below which a given % of data lies.
- **IQR** — Q3 − Q1, the spread of the middle 50%.
- **Outlier** — a value far from the rest.

Remember this: Always report a centre *and* a spread. Use the median when outliers are present, and use SD or IQR to show how varied the data is.

Name: _____ Date: _____

Activity

Task 1

Use the `income` array. Remove the manager's salary — how much do the mean and median change? Explain why.

Task 2

For `class_a` and `class_b`, write one sentence each describing the class using mean and SD.

Task 3

Use the IQR rule to check whether 2,500,000 RWF is an outlier in `income`.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. Which measure of centre is least affected by one very large value?
   A. Variance
   B. Range
   C. Mean
   D. Median
2. What is the median of 3, 9, 4, 1, 7?
   A. 4
   B. 7
   C. 4.8
   D. 9

3. What does the standard deviation describe?

A. How far values typically are from the mean
B. The largest value
C. The most common value
D. The number of values

4. Two classes have the same mean mark but class B has a much larger SD. What does that mean?

A. Class B's marks are more spread out
B. Class B has more students
C. Class B did better
D. Nothing

5. What is the IQR?

A. The mean of quartiles
B. Q3 − Q1, the spread of the middle half
C. Max − min
D. The median

6. Why should a report give a spread as well as an average?

A. To fill space
B. Spreads are always zero
C. Because an average alone can hide how varied the data is
D. Averages are always wrong

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